# VASYL SYNKO

## BACKEND DEVELOPER

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LinkedIn

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## SUMMARY

Backend engineer with 3+ years of experience in the fintech and blockchain domain. Built low-latency trading systems for cross-chain arbitrage on decentralized and centralized exchanges. Owned production microservices for staking, withdrawals, and exchange integrations – distributed sagas with compensation, idempotency under retries, and locking that keeps ledgers consistent when an external provider fails mid-transaction.

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## WORK EXPERIENCE

**Backend Developer, Cogitize** **January, 2026 – Present**

- Owned two production backend services end to end: a staking/earn microservice for a crypto exchange, and a multi-tenant game backend platform.
- Designed the staking service integrating a third-party exchange's investment products with the internal ledger and custody accounts – NestJS, PostgreSQL, Kafka, hexagonal architecture.
- Built a provider-agnostic engine for money-moving external calls: ambiguous timeouts and 5xx responses resolve through an idempotency-keyed status lookup instead of a blind retry, with a database unique constraint as the final backstop against double execution.

**Backend Developer, Peiko** **March, 2025 – November, 2025**

- Built backend services for two products – centralized crypto exchange platform and a social network with a built-in token economy (NestJS microservices in a monorepo, PostgreSQL/MySQL).
- Implemented the reward system awarding platform tokens for user activity, including balance accounting and the rules for what earns what.
- Implemented asynchronous transaction processing with BullMQ and RabbitMQ job queues, and optimised database schemas and queries for financial data

**Backend Developer, Peanut Trade** **August, 2024 – March, 2025**

- Built a real-time indexing pipeline for Solana DEXs: consumed block data over WebSocket from nodes and decoded on-chain program state (PDA, Borsh) to keep pool reserves and volumes current across venues.
- Ported AMM protocol math into the service to compute swap outputs locally instead of simulating through RPC, cutting per-route evaluation latency on the hot path.
- Implemented algorithms for constructing arbitrage routes across DEX venues and filtering unprofitable paths before submission.
- Built microservices for price analytics, swap tracking and strategy execution, with message-driven communication over Redis Pub/Sub.
- The system ran in production, executing trades with live capital.

**Backend Developer, Malevich** **June 2023 – April 2025 (full-time through Aug 2024, part-time maintenance thereafter)**

- Designed and built backend services in NestJS across microservice and monolithic architectures.
- Built and optimised PostgreSQL and Redis data layers, improving query performance and cache efficiency.
- Implemented GraphQL endpoints for flexible client data access and REST APIs for external integrations.

**Algo Trader & Dev, DeFi Tracker (startup)** **November, 2025 – March, 2026**

- Blockchain research (EVM & Solana chains).
- On-chain transaction parsing and event decoding.
- Working with crosschain bridges.
- Cross-chain arbitrage analysis & transaction processing (DEX–DEX).
- Market inefficiency & arbitrage research.

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## SKILLS

- Languages: TypeScript, JavaScript, SQL
- Backend: NestJS, Node.js, REST, GraphQL, gRPC, WebSocket, microservices, event-driven architecture, DDD
- Data: PostgreSQL, MySQL, TypeORM, Sequelize, Redis, schema design, query optimisation
- Messaging & async: Kafka, RabbitMQ, BullMQ, Redis Pub/Sub
- Blockchain: Solana (on-chain parsing, PDA, Borsh), Ethereum, Solidity, DEX/CEX integrations
- Infrastructure: Docker, Kubernetes, GCP, GitLab CI, GitHub Actions, Jest

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## EDUCATION

**Bachelor of Software Engineering – Uzhhorod National University**

- Graduated: July, 2025